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From bureaucratic machines to inter-organizational networks

Characterizing the response to the World Trade Center crisis

J. Ramon Gil-Garcia
University at Albany State University of New York, Albany, New York, USA and Centro de Investigacion y Docencia Economicas, Mexico City, Mexico

Theresa A. Pardo
Center for Technology in Government, University at Albany, SUNY, Albany, New York, USA, and

Djoko Sigit Sayogo
Center for Technology in Government, University at Albany, SUNY, Albany, New York, USA, and Department of Economics, University of Muhammadiyah at Malang, Malang, Indonesia

Abstract

Purpose – This paper aims to argue that the structure of the response to the World Trade Center (WTC) crisis can be characterized as an inter-organizational network and the majority of the activities can be identified as network management.

Design/methodology/approach – Using a mixed-method research strategy encompassing in-depth interviews and a sociometric survey, the authors characterize the response as an inter-organizational network and describe significant factors that facilitate the effective functioning and management of an emergency response.

Findings – The results provide empirical support for the claim that the management of the WTC response was very different from normal government operations in many respects. However, it was also found that complete detachment of the network-form of organization from bureaucratic hierarchy is not always possible in an emergency response, particularly in terms of leadership and the availability of resources.

Originality/value – The authors argue that bureaucratic leadership exists in political layers and is sometimes needed to provide social value to the general public and promote their engagement. Finally, the authors found evidence that the effectiveness of networks in an emergency response is influenced by certain enabling conditions, such as the severity of events, and suggest some implications for government operations.

Keywords Bureaucracy, Emergency response, Bureaucratic hierarchy, Inter-organizational networks, Network governance, World Trade Center crisis

Paper type Research paper

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1. Introduction

Disaster and emergency response increasingly involves coordination of multiple organizations across different levels of government, as they handle incidents that are beyond the normal capability and jurisdiction of any single organization (Curnin *et al.*, 2015). The World Trade Center (WTC) emergency response in 2001, for instance, involved – directly and indirectly – a total of 1,607 organizations, including 73 federal organizations (Kapucu, 2006). For situations like this one, some scholars argue that a dynamic and flexible network structure is more favorable than the traditional top-down, command-and-control bureaucratic model (Waugh and Streib, 2006; Moynihan, 2009). Thus, recent research has begun to study the factors that determine the effectiveness of network governance in an emergency response, including leadership (Donahue, 2006; Waugh and Streib, 2006), awareness building and participation (Comfort, 2005; Kiefer and Montjoy, 2006), flexible guidelines and rules (Donahue, 2006), scenario-based planning (Comfort, 2005; Donahue, 2006) and coordination and trust relationships (Kapucu, 2006).

Disasters have become increasingly regular (Curnin *et al.*, 2015). The International Disaster Database[1] shows that the number of disasters that occur each year across every continent rose significantly in the 1990s and 2000s. Average annual damages from such disasters are staggering; for instance, total losses from climate- and weather-related disasters in 2011 alone are close to US$200bn (Smith and Katz, 2013). Analysts estimated approximately $3m in damages from US weather-related disasters alone in the early part of 2015[2]. Such disasters not only result in significant economic losses but also immeasurable human suffering. Effective emergency response could minimize potential losses and suffering inflicted by these disasters. Thus, an understanding of the critical factors and lessons learned from a successful response could contribute to the development of a more effective emergency response.

This paper characterizes an emergency response as an inter-organizational network and describes critical factors that facilitate effective management of an emergency response, using the September 2001 WTC crisis as the case. It specifically addresses the following questions:

Q1. How is an effective emergency response organized and managed?

Q2. What are the critical factors affecting effective management of emergency response?

Q3. Are some of those characteristics related to a network form of organization?

The WTC response provides a rich research setting to understand factors that contribute to the effectiveness of an emergency response; in this paper, we highlight lessons learned, which remain salient today more than a decade after the crisis. As conveyed in a newsletter from the Center for Technology in Government, the 2001 WTC response:

> […] provides a unique and unrivaled research setting for increasing our understanding, and measuring the value of, collaboration, and essential information- and technology-intensive policies and functions of government[3].

Based on rich interview data and a sociometric survey, we suggest that the dominant organizing form of the WTC emergency response resembles a network structure. We adopt Powell's (2003) argument and define network structure as a form of social exchange based on relationships, mutual interests and reputation. To support our proposition, we critically examine results from interviews and a sociometric survey by contrasting the findings to the characteristics of bureaucratic hierarchy. As argued by Tirole (1986), hierarchies contain vertical structures with cascading layers that connect the principal and the agent. In this paper, we adopt Weber's (1946) definition and specifically refer to hierarchy as bureaucratic hierarchy; a hierarchical structure that is governed by formal rules and regulations.

We present our arguments based on the main characteristics of network structure and assess the extent to which these characteristics contributed to the success of the response efforts. This paper is organized in six sections including the foregoing introduction. Section 2 presents a review of the literature that systematically compares hierarchies and networks, emphasizing some of the main characteristics of networks as an organizing form. This section attempts to characterize bureaucratic hierarchies and networks as distinct forms of organization and contrast their main features. Section 3 delineates the research design and methods and includes a brief description of the data collection and analysis procedures. Section 4 presents the results of analysis of both the interview data and the sociometric survey using several social network analysis techniques. Section 5 presents our discussion of the results, theoretical and practical implications of the study and a few areas for future research on this topic. Finally, Section 6 provides some concluding remarks.

2. Understanding hierarchies and networks as organizational forms

Rich discussion of alternative forms of organizing started with Coase's (1937) seminal paper *The Nature of the Firm*, where he asserts that command systems were sometimes better than price mechanisms (Coase, 1937, p. 387). Williamson (1975) revisits the concept of organizational forms and proposes a dichotomy between market and hierarchy as the only possible forms of organizing. Moving from market to hierarchy as organizing forms depends on the frequency and build-up of knowledge and the frequency and nature of transactions. For instance, transactions will be governed under hierarchy when they occur frequently and thus require specific investments (Williamson, 1975). The discussion becomes broader when Granovetter (1985) challenges this dichotomy by expounding the importance of social relations. Powell (2003, p. 300) subsequently argues for a form of organizing that neither represents market nor hierarchy, has its own logic and is governed by "certain forms of exchange [that] are more social–that is more dependent on relationships, mutual interests, and reputation – as well as less guided by a formal structure of authority", later coined as network organization.

Scholars have since elucidated elements that distinctively characterize the three forms of organizing. In contrast to the other two, the market view does not propose a specific structure as a form of organizing and economists generally view the fluidity of price as the organizing mechanism (Williamson and Winter, 1991). Hierarchy and network, on the other hand, suggest certain structures that characterize each of them as a particular organizing form. Scholars argue that hierarchy is a form of organization characterized by the employment of graded authority to influence resource allocations in the organization where the higher offices supervised lower ones (Weber, 1946, pp. 196-197). This simplified structure is particularly efficient to mitigate complexities due to "the limits of human intellective capacities" (March and Simon, 1958, p. 169).

Hierarchy is efficient in reducing organizational complexity because it simplifies tasks and authority into a clear and graded structure, and thus, it reduces the cognitive load of individuals. In contrast, network as a form of organization is characterized by flexibility and adaptability to the complexity of environmental changes (Fulop, 1996) and particularly apt for circumstances that require efficient, reliable information (Powell, 2003).

Building from a review of the existing literature studying bureaucratic hierarchy and network structure, this paper identifies and compares ten features that characterize network structure as an organizational form. We use these ten features to understand the WTC emergency response and assess to what extent the response was a network and whether these characteristics were important for a more effective and successful response. Considering the dichotomy between network and hierarchy in terms of structure, we argue that if the response does not fully conform with the ten features of network structure, then the differences could represent some characteristics related to bureaucratic hierarchy.

We also propose that the characteristics of a network structure can be categorized into four important aspects of organizing: generating goals, managing functions, setting the structure and promoting coordination and communication. The first step of organizing is generating organizational goals, which in the case of a network, are framed through the alignment of members' goals. The goals will be implemented through sets of management functions in forms of leadership and authority and the development of trusted relationships. The management of functions will be organized in a particular structure that defines the division of tasks and deployment of resources. In the network, the structure is characterized by lean, flat and dynamic structure and flexible roles and responsibilities. Often, network structures need redundancy in terms of rules and requirements. Their flexibility frequently needs the relaxing of rules and the bypassing of red tape to achieve the network goals. Finally, the communication and coordination of tasks is crucial in a network structure. In a loosely coupled organizational structure such as a network, it is crucial to maintain the collaborative environment. Hence, decision making is performed taking into account the different views of network members. Boundary objects and the role of boundary spanners are crucial in accommodating the different views and in communicating and coordinating the tasks. Often in a time-constrained situation such as an emergency response, communication and coordination are strengthened and accelerated through the co-location of network members. Below we provide a brief description of each of the ten network features found in the literature. They are organized in the four categories mentioned above. A summary of the review is presented in Table I.

2.1 Generating goals

2.1.1 Framing shared goals. As a network structure relies on connecting loosely coupled relationships, the development of a common purpose becomes a prerequisite (Chisholm, 1996), such that framing goal alignment among members is a crucial function of network managers (Agranoff and McGuire, 2001). This is important because many times actors are collaborating for the first time or in a new context. In addition, as formal authority develops gradually (Moynihan, 2009), networks use goal alignment to establish authority and legitimacy (Rethemeyer and Hatmaker, 2008). Therefore, for

Table I. Comparing hierarchies and networks

Attribute	Hierarchy	Network	References
Generating goals			
Framing shared goals	Individual goals and interests are preserved and forced to align with organization goals	A distinctive character of a network as compared to hierarchy is in development of collective purposes Framing collective purposes is necessary to forge authority and legitimacy, provide motivation for members and convince the stakeholders to participate	Chisholm (1996), Agranoff and McGuire (2001), Moynihan (2009). Rethemeyer and Hatmaker (2008)
Managing functions			
Leadership and authority	The locus of leadership is clear in the hierarchy The cascading delegation of authority is pre-defined by its leader The leader's role is to supervise Leadership resides in one individual	Leadership is based on social relations and situations and often obscure The structure of a network influences the emergence of a leader; thus, leadership is conceptualized as influence over processes The leaders' roles are as brokers and mediators There could be multiple leaders in a network. Leadership is shared or distributed	Brass *et al.* (2004), Balkundi and Kilduff (2006), Daly and Finnigan (2011), Sutanto *et al.* (2011), Carson *et al.* (2007), Weber (1946), Curnin *et al.* (2015)
The role of trust	Trust is low due to the standardization and codification of rules in hierarchy Interpersonal trust is diminished through the administrative fiat	A higher degree of trust is needed to curb possible opportunistic behavior Affective-based trust is important for partner selection and retention	Thompson (2003). Granovetter (1985), Van Alstyne (1997) Rousseau *et al.* (1998), McKnight et al (1998), Lewis and Weigert (1985), Rethemeyer and Hatmaker (2008)
Setting the structure			
Lean and flat organization structure	Hierarchy in bureaucratic governance represents a steep pyramid structure where a central unit is supported by many levels	Network governance has a flatter structure due to the elimination of middle layers	Leibenstein (1987), Van Alstyne (1997), Podolny and Page (1998)

(*continued*)

Attribute	Hierarchy	Network	References
Flexibility in roles and responsibilities and dynamic structure	Hierarchy employs standardization and task specialization Formal relations and authority dominate hierarchy The boundary is fixed, rigid and definite and involves stable ties	Informal relations and persuasive authority are dominant in a network The boundary of a network is flexible, permeable and dynamic	Binz-Scharf (2003), Van Alstyne (1997), Agranoff and McGuire (2001)
Regulatory environment and bypassing red tape	Law is the source of rationality and the primary virtue is reliability Highly regulated environment	Culture is the source of rationality and the primary virtue is flexibility Authority and legitimacy is negotiated A network responding to a major disaster might bypass many formal rules to directly respond to a given situation	Considine and Lewis (2003), Koontz and Thomas (2006), Van Alstyne (1997), Benini (1999).
The role of redundancy	Redundancy is equated with inefficiency, hence avoided	A network has a less definite timeframe and no general expectation of permanence Network forms often actively built in some redundancy	Fulk and DeSanctis (1995), Thompson (2003), Semlinger (2008)
Promoting coordination and communication			
Collaborative decision-making	Centralization of decision making and authority Communications are persistent, through formal channels, and in the forms of one-to-many (or many-to-one) Top-down decision making	Decentralization of authority and decision making The communication pattern is many-to-many and less formalized Collegial nature of decision-making process	Binz-Scharf (2003), Waugh and Streib (2006)
Boundary objects and boundary spanners	Boundary spanners are not preferred in hierarchy where roles and positions are clearly defined	A boundary object is needed to bridge the differences between members, while boundary spanners act as brokers to facilitate intergroup transactions	Pawlowski *et al.* (2000), Callister and Wall (2001); Richter *et al.* (2006)
Importance of co-location	Hierarchy and co-location offer potential financial benefits through resources pooling and sharing, such as sharing administrators and office infrastructure	Co-location increases the likelihood and ease of interaction, which enhances network governance ICT could facilitate distant connections and invoke a "perceived proximity"	Jones *et al.* (1997), Cannella *et al.* (2008), Polzer *et al.* (2006), Wilson *et al.* (2008), Eggs and Englert (1999)

Table I.

networks, generating common goals is very important and a way to start forming and consolidating the network as a whole and the relationships among members.

2.2 Managing functions

2.2.1 Leadership and authority. Bureaucratic hierarchy favors leadership built upon formal authority with a cascading delegation of authority along the hierarchical ladder (Weber, 1946). In a network structure, leadership is transformational; it is obscure (Balkundi and Kilduff, 2006) and shared or distributed among the relationships (Carson *et al.*, 2007). Leadership is not appointed but emerges from its structure and is affected by the structure itself (Balkundi and Kilduff, 2006; Brass *et al.*, 2004; Sutanto *et al.*, 2011). Meaning, while its formation constitutes a contested process that is influenced by its members (Brass *et al.*, 2004), the position of a member in a network structure influences the selection of the leader. For example, a central actor[4] is more likely to be chosen as leader, as a central position is associated with higher power (Brass *et al.*, 2004). Leaders in a network assume various roles depending on the network structure, as brokers (Daly and Finnigan, 2011), boundary spanners (Balkundi and Kilduff, 2006), mediators (Sutanto *et al.*, 2011) or liaisons (Curnin *et al.*, 2015).

2.2.2 Role of trust. Trust is the backbone of a network structure (Thompson, 2003); it sustains collaborative actions (Granovetter, 1985) and curbs opportunistic behavior (Van Alstyne, 1997). Trust matters in a network structure because the governance of a network is not based on administrative fiat through rules, systems and structure (Van Alstyne, 1997). Formal authority develops gradually in a network (Rethemeyer and Hatmaker, 2008). As argued by Rousseau *et al.* (1998), trust is a multi-dimensional concept and the development of trust through the formation of relationships is influenced by the interchange of affective and cognitive factors (Lewis and Weigert, 1985). For example, trust might be influenced by cognitive aspects such as reputation, credentials and affiliations at the initial stage of building relationships because of missing affective factors such as personal affection (McKnight *et al.*, 1998).

2.3 Setting the structure

2.3.1 Lean and flat organizational structure. While bureaucratic hierarchy has a steep structure with many intermediate levels (Leibenstein, 1987), a network has a flatter structure due to fewer intermediate layers (e.g. fewer middle managers, levels or authority and oversight). The lean and flat structure is necessary for maintaining collegial coordination and participative decision-making (Van Alstyne, 1997; Podolny and Page, 1998).

2.3.2 Flexibility in roles and responsibilities and dynamicity of structure. In bureaucratic hierarchy, roles and responsibility are governed by task specialization and rules and procedures (Binz-Scharf, 2003), with fixed task and organization boundaries based on authority and stable ties (Binz-Scharf, 2003; Van Alstyne, 1997). Roles and responsibilities are flexible and dynamic in network structures and they are governed by the process of activation, deactivation and mobilization of ties and connections (Agranoff and McGuire, 2001). Consequently, structure and boundary in networks are flexible, permeable and relative as shaped by dynamic ties (Van Alstyne, 1997).

2.3.3 Regulatory environment and bypassing red tape. The governing of bureaucratic hierarchy is achieved through law, rules and regulations in which deviation from them is strictly discouraged (Koontz and Thomas, 2006). The virtue of a network structure, on the other hand, is flexibility (Considine and Lewis, 2003; Powell, 2003). Tasks in a network structure are project-oriented and thus often the organizational structures are not permanent and stable over time (Van Alstyne, 1997) with the objective to get the job done. In situations such as an emergency response, the objective to get the job done often prevails over formal rules. Responders, dictated by urgency, occasionally bypass red tape to the benefit of time-sensitive efforts (Benini, 1999).

2.3.4 The role of redundancy. As the structure of bureaucratic hierarchy is built upon rigid formal rules and regulations, bureaucratic hierarchy equates redundancy as a deviation from rules and as contributing to inefficiency (Thompson, 2003). The merit of a network structure, on the other hand, is flexibility. As a result, built-in redundancy is often a necessity in the network to avoid the problems of overload, vulnerability (Fulk and DeSanctis, 1995) and "lock-in" – a situation where a network structure cannot expand because of strong ties or full connections (Semlinger, 2008). A certain degree of built-in redundancy is necessary to provide an entry point for potential members or exit options for existing members so that the network can maintain flexibility (Semlinger, 2008). It is also crucial to factor in redundancy in creating extreme event-ready resilient information infrastructure (Scholl and Patin, 2014).

2.4 Promoting coordination and communication

2.4.1 Collaborative decision-making. Hierarchy favors a centralized structure where authority and information flows vertically in cascading forms from the top level to the bottom (Binz-Scharf, 2003). While such a structure permits decisiveness in the decision-making process, it lacks flexibility and collaborative processes. In an emergency response involving multiple organizations, lack of flexibility can hinder effective relief effort. As argued by Waugh and Streib (2006), goal conflicts are very common in disaster and emergency response. A network structure permits flexible and participative decision-making. In multi-organizational settings such as an emergency response, where each organization might have different goals and agendas, flexibility through the process of participatory decision-making could minimize conflicts. Collaborative partnerships are essential for the acceptance of changes and could support more targeted and effective activities in the case of emergency (Higgins *et al.*, 2015).

2.4.2 Boundary objects and boundary spanners. Boundary objects and brokers are seen as the two important tools to facilitate coordination, particularly in a network structure (Pawlowski *et al.*, 2000). A boundary object is necessary for bridging communication across different boundaries to create shared and mutual understanding (Pawlowski *et al.*, 2000), a vital aspect in framing shared goals. Broker or boundary spanners are imperative in brokering intergroup transaction and managing intergroup conflicts (Callister and Wall, 2001; Richter *et al.* 2006).

2.4.3 The importance of co-location. Although information and communication technologies (ICTs) could facilitate interchanges and communication across geographically dispersed locations (Wilson *et al.*, 2008), geographical proximity enhances the effectiveness of a network (Jones *et al.*, 1997). The importance of co-location, or geographical proximity, increases when uncertainties in the environment intensify (Cannella *et al.*, 2008). Co-location increases the likelihood and ease of network interactions (Jones *et al.*, 1997); encourages the development of trust and inhibits possible conflicts (Polzer *et al.*, 2006); and improves the competitive ability of a network (Eggs and Englert, 1999).

3. Research design and method

This study uses a mixed-method research strategy, combining both quantitative and qualitative approaches. More specifically, we complement the results from in-depth interviews with results from a socio-metric survey. The use of multiple methods allows triangulation and cross-validation of results. Multi-method approaches "neutralize or cancel the biases of other methods" (Creswell, 2009, p. 15). By using multiple methods within a single study, the weaknesses of individual methods can be identified and partially solved (Gil-Garcia and Pardo, 2006). This mixed methodology also enables us to both understand the structural characteristics of crisis response governance, mainly through the social network analysis, and identify the underlying drivers for the emergence of this structure in crisis response, mainly through the interviews. Overall, a multi-method strategy provides a more complete understanding of the characteristics of the response as an organizational form and the importance of those characteristics to the success of the response. Whenever possible, we provide evidence from the two methods in each variable.

3.1 Data collection

3.1.1 Semi-structured interviews. We conducted face-to-face semi-structured interviews and one video interview. This approach enabled researchers to gain understanding of emergency responders' experiences, focusing on the geographic information systems (GIS) team, a sub-network within the larger network of responders involved in the WTC crisis response. Researchers from [*Deleted for peer review process*] conducted interviews with 29 first responders in 2002 and 2003. We also analyzed a video interview with prominent members of the response team in the command center at Pier 92. Pier 92 was the location of the emergency operation center (OEC) during the response to the WTC crisis, where the GIS team, among other responders to the crisis, was located (Harwood, 2001). Pier 92 was selected as the focus of data collection activities because it was the center of the response efforts. Interviewees were team members of the GIS sub-network and came from a variety of different professions and affiliations, including government agencies, non-profit organizations, consultants, academics and private firms. Interview subjects were selected based on the roles they played in the response and how they were recognized by others as instrumental. The diversity of the backgrounds of the interview participants enabled us to build an in-depth understanding of the response situation by comparing interviewees' different perspectives and roles during the WTC crisis.

3.1.2 Sociometric survey data collection. A sociometric survey was also conducted during the interview. At the end of each interview, interviewees were asked, "Please name 5 individuals who were instrumental in resourcing, facilitating, managing, leading, supporting, or inspiring your response efforts". Based on the interview results, we created a one-mode network of reputation-based relationships. The one-mode network represents person-by-person relationships. If person i identifies person j as instrumental in the WTC crisis response, then we assign a value of 1 (and 0 otherwise). This research did not symmetrize the relationship; if actor i says that actor j was instrumental during the response, it does not automatically indicate that actor j will think the same thing about actor i.

3.2 Analysis techniques

This research uses two complementary analytical approaches:

(1) qualitative analysis based on semi-structured interviews; and

(2) social network analysis of the relationship pattern within the GIS sub-network.

Information provided during the interview is used to perform a preliminary analysis of the network characteristics behind the successful WTC response. Some direct quotations are used to provide evidence in terms of the network characteristics of the WTC crisis response. Social network analysis of the relationship structure within the GIS sub-network is used to provide better understanding of how the social structure in the WTC crisis response influenced the effectiveness of the response. The social network analysis presents a visualization of the network structure and related measurement routines, focusing on the cohesion, centrality, core/periphery and the ego-network brokerage measurements.

4. Analysis and results

The WTC emergency response is regarded as a success story of a collaborative network where scores of organizations across different sectors, jurisdictions and levels of government worked collaboratively in the relief efforts (Kapucu, 2006; Moynihan, 2009). Based on content analysis of news and other publications in the 21 days after the attack, Kapucu (2006) identified 1,607 organizations that were involved in the response, including 73 federal organizations. Judging from the number of organizations involved and the ways they were coordinating, it is reasonable to argue that the WTC emergency response typifies a network structure as opposed to a bureaucratic hierarchical structure. Based on the interviews and sociometric survey with the GIS response team, we provide evidence in the following two sections that the response had a network structure. The first section describes the network structure of the GIS team based on the sociometric survey. The second section describes the organizing form of the response where we combine results from our in-depth interviews with the results from the sociometric survey to strengthen and cross-validate the arguments.

4.1 Overview of the network structure: a GIS sub-network perspective

We analyzed the interpersonal coordination and communication of the GIS team. Our social network analysis results confirm and enrich the findings from Kapucu (2006) in that we found that the GIS team had diverse and heterogeneous members consisting of actors from different organizations and government levels as indicated from the visualization (Figure 1) and the density values (Table II). This resulted in less cohesive and less dense relationship patterns with low density values of 0.0554, a high standard deviation of 0.2288 and low compactness of 0.161.

These results indicate a sparse network which, as argued by Walter *et al.* (2007), could lengthen communication channels, create the possibility of communication distortion and pose a coordination challenge. However, these challenges were not an issue for the GIS team, as the results indicate; the GIS team resembles a core-periphery network structure. This means that the GIS team's structure has a strong core with a cohesive sub-group that connects sparse and unconnected nodes in the periphery. In general terms, the structure of the GIS team is due to the combination of the pre-existing NYC GIS members as the core and the "newcomers" added later to the team as the periphery.

In terms of the centrality of actors, we identified contrasting results from Kapucu's (2006) findings. We found that the GIS network has low in- and out-degree centrality with a value of 1.77, meaning that, on average, the most central actor only receives and gives two ties with other members. In other words, the network members saw themselves as equals in terms of power, influence and conveyance of information. On the other hand, we found high variance of betweenness centrality of 1,543.98, meaning that there are few actors who were skilled in acting as boundary spanners or liaisons connecting members to one another. We found a subset of several actors with high brokerage value (Table II) in their role as liaison.

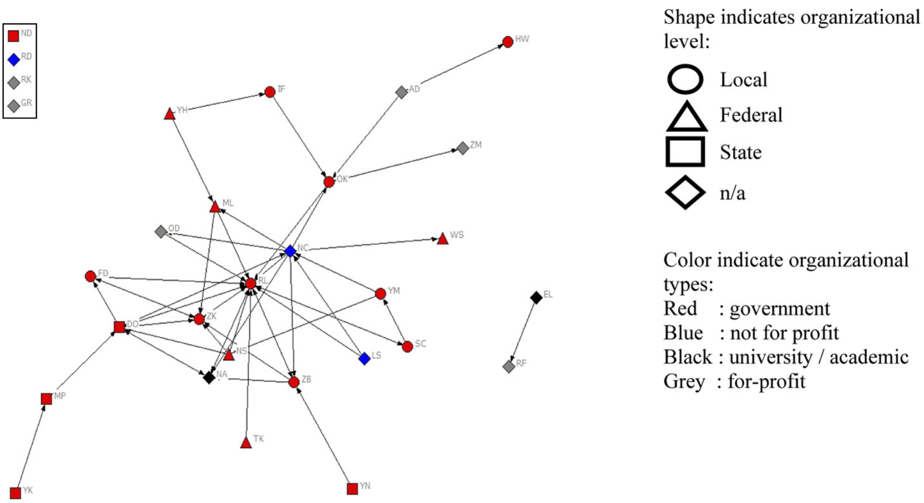

Figure 1. Visualization of network based on attributes

Table II. Density, centrality and brokerage measurement results

	Centrality measures			
Measurement	Mean	SD	Variance	Centralization (%)
Out-degree centrality	1.767	1.802	3.246	22.24
In-degree centrality	1.767	2.642	6.979	40.07
Betweenness centrality	18.567	39.294	1,543.98	19.29

Density measures		
Measurement	Mean	SD
Density	0.055	0.229
Core density	0.500	
Periphery density	0.020	
Compactness	0.161	
Average distance	2.793	

Brokerage measures	
Actors	Liaison
RL	58
OK	6
DO	14
NC	20

Source: UCINET results of WTC sociometric data

4.2 Characterizing the response to the WTC crisis

In this section, we discuss the management of the WTC emergency response in terms of the ten features characterizing network structure, as identified in the literature review. When appropriate, we use and combine the results from the interviews and the results from the social network analysis to strengthen our arguments.

4.2.1 Framing shared goals. The GIS network structure indicates sparse connections among the actors involved in the team (Table II and Figure 1). The sparse connections might indicate distant relationships and a higher possibility of conflict. In such situations, prompt development of shared goals diminishes the risk of conflict. Subsequently, adopting participative and collaborative approaches is more desirable than bureaucratic hierarchy, considering that the actors with sparse connections might have different agendas and political settings. Swift development of agreed-upon goals and curtailing individual political agendas can lead to efficient management of response. In the WTC case, the possible damaging effect of politics was mitigated by the gravity and severity of the crisis. As one of the interviewee points out:

> We're dealing with really bad things that have happened [...] . People are much more willing to do their best to help the situation [...] . There was no politics.

4.2.2 Leadership and authority. Our analysis identifies two types of leadership that manifested at two different ways; political and operational. First, the leadership role assumed by the NYC Office of the Mayor existed at a political level. This leadership came from the bureaucratic hierarchy and their role was to provide the overall vision for the response and to reassure the public about the progress of the relief activities. As indicated by one interviewee "They [Office of the Mayor] really calmed the city and they gave people a faith that we could get through this". The Mayor, as the highest level of hierarchy for NYC, was recognized as the ultimate leader who served to ensure the continuity of the relief response network. We also found the existence of leadership at the operational level in the form of network-based leadership. As the interview results show, this leadership is fluid and is generated from expertise and know-how. We found that the role of leaders at the operational level was to provide resources, to bridge connections as boundary spanners and to activate (or add) relationships to the network. The network leadership was focused on getting the job done. For instance, one of the interviewees from Citywide GIS for the NYC DoITT described how the GIS Manager from the city's Department of Parks and Recreation immediately delivered computer terminals and networking capability to build the infrastructure needed to support the emergency mapping center.

4.2.3 The role of trust. Our interview results illuminate the significant role of trust in supporting the success of the WTC emergency response. We identified that both forms of trust – affective and cognitive – played an important part in the response. Cognitive-based trust emerged in the political setting of the response, particularly for the relationship between the Office of the Mayor and the public. Cognitive-based trust manifested in terms of the reputation and bureaucratic position of the Mayor and was necessary to assure the public that there was continuous support for the response. It is particularly challenging to achieve a strong sense of public security using a network structure because the public often seeks formal and legitimate assurance that can best be provided by political leadership and bureaucratic hierarchy. At the network level, cognitive-based trust is apparent in the use of expertise to negotiate relationships. The network structure used expertise as a frame of reference to activate new actors in the network. We also found evidence of affective-based trust that facilitated coordination and decision-making processes. This affective-based trust manifested in two forms: the history of relationships and the culture of the GIS community of practice. An interviewee from Citywide GIS for NYC DoITT indicated:

> It's just in the nature of people who work with geographic information, that we just naturally coalesced together and worked cooperatively and shared data; maybe it's our instinct to do that.

The sharing and cooperative culture in the GIS community of practice drove the creation of affective-based trust in the governance of the response. The interviews also point to the cultivation of prior individual relationships as the basis for trust creation in the network. As mentioned by one of the interviewees, the response was "just the wide network of people and friends and associates that we had cultivated over the past seven years […] [and we] all came to play".

4.2.4 Lean and flat organizational structure. Our social network analysis results clearly show an emergency response network with a flat structure with few intermediate layers. We found the existence of a core-periphery structure with high core density of 0.500, high average distance of 2.793 and low periphery density of 0.020. We also found in- and out-degree centrality of 1.77 which, after considering the numbers of nodes (30 actors), signifies low centrality. This low centrality indicates a flat structure with little power differences among the actors. There was little or no hierarchy in terms of power and influence among the actors comprising the GIS network.

4.2.5 Resiliency of roles, responsibilities and structure. Our interview results indicate that the management of the response demonstrates a higher degree of flexibility in terms of roles, responsibilities and even structure. Roles and responsibilities were not formed based on formal authority but mostly based on needs and expertise. The need to get the job done superseded the bureaucratic disposition of the actors in the WTC emergency response. The interviewees alluded to the case where a GIS Manager from New York changed his role to being a logistics provider due to his local knowledge and the need of the network to get the job done. The interview results also indicate how dynamic the network structure was in terms of activation (or engagement of new members) and mobilization (the acquisition of resources). Activation and mobilization in the WTC response was accelerated by the need to fulfill particular tasks based on the expertise of selected participants. For instance, one of the interviewees mentioned:

> When we identified the need for […] thermal [images] of the site, they immediately added that to the roster of services and they too became a participant from 128 miles away, up in Albany.

4.2.6 Regulatory environment and bypassing red tape. In concurrence with extant literature (Dawes *et al.*, 2004), our interview results indicate that in an emergency response, rules and regulations are often negotiated in exchange for rapid delivery. There is willingness to forgo or relax routine procedures and circumvent red tape for the benefit of achieving immediate relief or certain outcomes in the response process. The interview results clearly show that critical needs dictated immediacy of response and helped to overcome bureaucratic red tape. The important thing in an emergency situation is to get the necessary work done. For instance, the GIS team built the emergency mapping center from the scratch and everyone contributed and brought whatever materials they had available. As one of the interviewees, then the GIS Manager from the NYC Department of Parks and Recreation, pointed out:

> One of the big issues at the first emergency operation center at the police academy was we didn't have a printer that was capable of supporting the map production. So I went up to my office and came down with the printer from my office.

Less clear, however, is what impact bypassing bureaucratic red tape had on the accountability of resources in the long term.

4.2.7 The role of redundancy. The case of the WTC response demonstrates the resilient properties of a network structure – the decentralized structure and lateral power relations – that can adapt to disruption and interference. Conversely, the case indicates the limitations of a bureaucratic hierarchy to adapt in the same way to disruption. The leaders of the Port Authority of New York and New Jersey Police Department (PAPD) perished in the collapse of the WTC tower during the crisis response. However, the loss of these leaders did not obstruct the response efforts. Our interview results show, and as denoted by Dawes *et al.* (2004), that due to the decentralized structure and lateral power relations in PAPD, other leaders quickly filled the void and carried the relief efforts forward. As indicated in Table II and Figure 1, decentralized and lateral relations are also manifest in the GIS network structure, which has sparse and diverse connections. The low overall density (0.0554), the high standard deviation of density (0.2288) and the low compactness (0.161) indicate a less cohesive structure.

4.2.8 Collaborative decision-making. Our analysis results suggest the significance of collaborative decision-making during the WTC crisis response. Collaborative decision-making is necessary to make coordination more efficient. There are two factors that influenced the success of collaborative decision-making in this case, namely:

(1) the culture of the GIS community of practice; and
(2) the affective-based trust built from previous relationships.

The interview results indicate the existence of a strong culture of cooperation and coordination embedded in the GIS community of practice. As stated by one of the interviewees from Citywide GIS for NYC DoITT, it is customary for the people in the GIS community of practice to work cooperatively and share data, so this almost becomes a norm for them. This cooperative culture became the key element in making collaborative decision-making possible. Affective-based trust was built upon years of relationships and experience working together, as described in our discussion of trust above.

4.2.9 Boundary objects and boundary spanners. Agranoff and McGuire (2001) suggest the importance of synthesizing[5] in the management of a network structure, in particular for coordinating and integrating the various network members. Our findings from the interviews and sociometric survey indicate the creation and use of two significant tools for synthesizing the network, namely, boundary spanners and boundary objects. Our analysis identifies the use of GIS maps as one of the boundary objects that facilitated communication and coordination within and beyond the network. One of the interviewees indicated how GIS maps helped to create shared understanding even in hectic situations: "Even in the chaos of the early moments, because we had the base [GIS] map, it greatly facilitated our response and ultimately our ability to communicate with each other". This finding complements the argument put forward in the paper by Harrison *et al.* (2007) on transformation of GIS into a dynamic decision-making tool in emergency response. Our finding further elucidates the role of boundary objects in helping the responders bridging the differences among them and improving the ability for the responders to communicate with one another.

In addition, coordination and communication were also significantly facilitated by the emergence and existence of boundary spanners. Boundary spanners are members of an organization who bridge the boundary of their organization to connect to the larger external environment (Burt, 1995; Williams, 2002). The results from our social network analysis show a high mean of betweenness centrality (18.567) and high variation (1543.89) in the betweenness centrality (Table II), which signifies that a few actors were very effective in acting as boundary spanners. In addition, our egocentric analysis also identifies several responders with high brokerage value, particularly in their roles as liaisons to bridge and connect the GIS network with the larger response network (Table II). The interviewee cited earlier from Citywide GIS for NYC DoITT described the emergence of a boundary spanner in the story about how the GIS Manager from the Department of Parks and Recreation used his local knowledge to identify the needed critical resources for the network and acted as a liaison. The GIS Manager from the Department of Parks and Recreation described his roles in his own words:

> Yeah, I was scrounging. One of my roles was people would tell me, "Hey, Paul, we need this off" or "Hey, Paul, we need this piece of hardware". I'd say, "O.K." and I'd get it.

4.2.10 The importance of co-location. Our results also identify the importance of having a physical center located in the same vicinity of the crisis, a co-location for responders. The emergency mapping center for the WTC crisis response was initially set-up at the Police Academy, but was later moved into the emergency command center (EOC) at Pier 92. Co-location at the EOC facilitated sharing among the responders. As one of interviewees stated:

> If we would have been co-located [from the beginning], it would have been easier for us to share […]. There are things that I would have been able to easily help them with that they had to learn on their own.

This excerpt signifies that co-location was crucial for two objectives. First, co-location facilitated the sharing of knowledge, information and expertise among responders. Second, co-location made it easier for the responders to educate others about the situation. Plausibly, co-location can lessen the learning curve and potentially accelerate certain response activities.

5. Discussion and implications

This paper describes how diverse actors managed an emergency response. We posit that the 2001 WTC emergency response demonstrates the use of both bureaucratic hierarchy and network structure into organizing a response. We present a summary comparing the use of a network structure against a bureaucratic hierarchy in Table III. This section discusses findings and provides some theoretical and practical implications of the work.

Attribute	WTC response	H	N
Framing shared goals	The gravity of the collective goals leads to development of shared authority and avoidance of conflicts		✓
Lean and flat organization structure	The low in- and out-degree centrality indicates a flat structure with little or no hierarchy in term of power and influences among the actors comprising the GIS network		✓
Flexibility in roles and responsibilities and dynamic structure	The structure, roles and responsibilities are built based on need and expertise to get the job done. The hierarchical position and roles are changed to fit with the need of the network		✓
Leadership and authority	The network is nested within the hierarchy. The network acknowledges the certain role of the political leader for public engagement. The political leader and network leader operate at different levels	✓	✓
Regulatory environment and bypassing red tape	There is willingness to forgo or relax routine procedures and skirt around red tape for the benefit of achieving immediate relief		✓
The role of trust	There is clear evidence of the use of both high levels of affective-based and cognitive-based trust, based on the historical relations and expertise	✓	✓
The role of redundancy	The response network is not dependent on a central figure for sustainability. There are several actors who presume similar roles		✓
Boundary objects and boundary spanners	The WTC crisis response uses both the boundary object (GIS map) and boundary spanners to ensure smooth coordination		✓
Importance of co-location	Co-location at Pier 92 helped smooth the coordination process, facilitating learning and sharing		✓
Collaborative decision-making	The low centralization measurement indicates that the work commitment in the sub-group is based on collegiality supported by a history of working together and culture/identity		✓

Table III. Comparing hierarchies, networks and the WTC response

5.1 Discussion

Using results from interviews with the responders of the WTC crisis response and sociometric survey data evaluating their relationships, we found evidence of the use of a network structure in every aspect of the response activities. Yet, we also found indications that the network structure was not completely detached from bureaucratic hierarchy, particularly in terms of leadership and resources. Thus, this study provides further support for the hybrid model of organization that combines hierarchy and network (Feiock and Jang, 2009). We also found support for the attachment of a network structure on a bureaucratic hierarchy (Rethemeyer and Hatmaker, 2008). Our findings further support Koliba *et al.*'s (2011) argument that the combination of formal authority and collaborative initiative, in fact, positively contributes to the effectiveness of network governance.

On the other hand, we also found the use of a network structure versus a bureaucratic hierarchy varies at different levels in emergency response. We found that bureaucratic hierarchy manifested at the political level and the network form was used at the operational level. While Rethemeyer and Hatmaker (2008) argue that the intersection of network and hierarchy is for resource fulfillment, we found that each had different functions. Having resources alone is not sufficient to manage and maintain collaborative efforts (Sorrentino and Simonetta, 2012). Bureaucratic hierarchy was primarily used for public-facing activities and strategies, whereas a network structure was used to get the job done. Our argument that network and bureaucratic hierarchy structures co-exist at different levels is clearly evidenced in terms of leadership. In the WTC response, the Office of the Mayor, supported by bureaucratic legitimacy, assumed the role of a stalwart champion and provided reassurance to the public during a difficult time. In concurrence with Caro's (2015) argument, the community needs reassurance when there is widespread distress. Although we differ slightly from Caro's opinion with regard to where this leadership role exists between network and bureaucratic hierarchy structures, we concur with Waugh and Streib (2006) that this leadership role requires bureaucratic support because the bureaucratic position is more permanent. The pressure for security during a disaster propels the public to seek assurance from the bureaucratic hierarchy (Waugh and Streib, 2006). The type of assurance from formal legitimacy and authority – as pointed out by Rethemeyer and Hatmaker (2008) – might not be readily available in a network.

Finally, in addition to the structure, the effective functioning and management of emergency response is significantly affected by certain enabling conditions. We argue that the enabling conditions influence the efficiency and effectiveness of emergency response. One of the enabling conditions is the gravity of the event. For instance, when the gravity of the event increases, the role of the bureaucratic leader in providing social value via engagement with the public becomes more efficient. The gravity of the event could also accelerate the formation of shared goals. The responders were more willing to work together and abandon their political agendas because the WTC situation was an extreme event; hence, shared goals developed more easily.

5.2 Theoretical and practical implications

Our results from both the interview and social network analyses enable us to put forward several possible theoretical and practical implications. This study found that a network structure cannot be completely detached from bureaucratic hierarchy even in a crisis situation. This finding supports the assertion of the mutual co-existence of a network and bureaucratic hierarchy structure for emergency response (Kapucu, 2006; Moynihan, 2009; Rethemeyer and Hatmaker, 2008). We posit that the connection of a network structure to a bureaucratic hierarchy manifests in the bureaucratic leadership and in the management of public communication. In emergency response, the intrusion of political leaders becomes necessary and does not hinder the operational progress when the specification of tasks is clear. We found evidence of clear distinction in the leadership role between bureaucratic hierarchy and network structures. While the political leader assumes the role of providing assurance to the citizens on the success of the response, this leader provides network managers at the operational level with more flexibility and discretion to get the job done.

We have to emphasize that such collaboration is possible when there is clear distinction and balance in the different leadership roles and when bureaucratic leaders provide network leaders with more leeway to innovate during the response. In such situations, the coexistence and even intrusion of bureaucratic hierarchy will not create conflict and disintegrate the cohesiveness of the network; instead, it supplements the effectiveness of a network. However, as argued by Garrett (2010), continuous and excessive exertion from the executive level to the network could stymie collaborative initiatives. Future research can study the roles of different leaders in response efforts and assess how their influence is related to their formal position and their level of involvement with other members of the network.

From a practical perspective, the clear distinction and balance of leadership in network and bureaucratic hierarchy structures is crucial to frame network authority and control within the network. For instance, Bate (2000) found that involvement in a network could make senior managers feel that they have to surrender their power, status and authority. When an emergency response is large and the complexity of organizing it is immense, there is increased likelihood of higher political control and intrusion from political figures who seek to control the network and prevent fragmentation (Davies, 2005). As we argue, higher political intrusions could instead result in disintegration of the network. Hence, we need a more effective way of framing authority and control in a network structure. Our findings point to the importance of the gravity of the event. The severity and impact of the event could ease the process of framing authority in the network. More research is also needed in terms of systematically comparing the roles of formal leaders in emergencies in which the gravity of the event is different and how this could impact the overall results of the response.

A final point worth making is that more attention should be placed on collaboration, not only for emergency response, but also for strategies and activities to prepare for those emergencies and to ameliorate their devastating effects. It is worth noting that with the rapid advancement of ICTs and the rich data collected by public and private organizations, a more systematic approach to decision making in emergency response could be possible. The current challenge is to better understand the critical factors and capabilities for creating such a systematic approach for timely and flexible decision making through collaborative activities and data integration, such as working collaboratively for data analytics in an emergency response. Future research should focus on this and other topics related to data, collaboration and data analytics.

6. Conclusion

We provide evidence in our paper that working collaboratively through the use of network structure for emergency response and disaster relief is a challenging endeavor and assessing the critical factors necessary for collaborative work could be crucial for a successful response. The results of our analysis indicate that the governance structure of the WTC crisis response resemble an inter-organizational network in which the functions and activities were mainly performed based on network management. However, our results also indicate that it is not always possible to have complete detachment from bureaucratic hierarchy, particularly in terms of leadership and resources allocation and deployment. The response to the WTC crisis could be characterized as an inter-organizational network, but more research is needed in terms of the intersections between hierarchies and networks as forms of organization. Instead of thinking about dichotomies or clear distinctions between different forms of organizing, in our case networks and bureaucratic hierarchies, a more useful perspective would be to understand the similarities, differences and intersections among different organizational forms and which combinations could be better alternatives for specific situations and contexts. Our paper is a first step in this direction and we hope other researchers use these initial ideas in the near future.

Notes

1. www.emdat.be/natural-disasters-trends
2. www.emdat.be/publications
3. www.ctg.albany.edu/publications/newsletters/innovations_2002_summer?chapter=1
4. A central actor is an actor with many ties to other actors in the network.
5. Synthesizing refers to the creation of conditions favorable for co-activities among the network members. For more discussion, see Agranoff and McGuire (2001).

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About the authors

J. Ramon Gil-Garcia is an Associate Professor of Public Administration and Policy and the Research Director of the Center for Technology in Government, University at Albany, State University of New York (SUNY). Dr Gil-Garcia is a member of the Mexican National System of Researchers as Researcher Level III and of the Mexican Academy of Sciences. In 2009, he was considered the most prolific author in the field of digital government research worldwide, and in 2013, he was selected for the Research Award, which is "the highest distinction given annually by the Mexican Academy of Sciences to outstanding young researchers". Dr Gil-Garcia is the author or co-author of articles in prestigious international journals in public administration, information systems and digital government, and some of his publications are among the most cited in the field of digital government research worldwide. J. Ramon Gil-Garcia is the corresponding author and can be contacted at: jgil-garcia@ctg.albany.edu

Theresa A. Pardo, PhD, is the Director of the Center for Technology in Government at the University at Albany, State University of New York, where she also holds Research Associate Professor appointments in Public Administration and Policy and Informatics. Dr Pardo serves as OpenNY Adviser to New York State's Governor Cuomo and is a member of the US Environmental Protection Agency's National Advisory Committee and the User Working Group of NASA's Socioeconomic Data and Applications Center, among others. Dr Pardo is founder of the Smart Cities, Smart Government Research-Practice Consortium and is Past-President of the Digital Government Society.

Djoko Sigit Sayogo is a Senior Lecturer in the Department of Economics and Business at the University of Muhammadiyah at Malang (UMM), Indonesia. Additionally, Djoko is the Director at the Center for Economics, Business and Entrepreneurships Development at the Department of Economics, University of Muhammadiyah at Malang. He is currently a Postdoctoral Associate at the Center for Technology in Government. He was working on the CTG team that aims to develop a proof of concept for global information systems using Semantic Web technologies based on OWL ontology, called I-Choose. He is a former Fulbright Presidential Scholar from Indonesia.